Answer Key & Teacher Notes

Session 1 — Graphing Ratios on the Coordinate Plane · Session 2 — Interpreting Points on Ratio Graphs

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Graphing Ratios on the Coordinate Plane

TALK PROMPTS

- What does the origin (0, 0) mean in the soccer-bag situation?
- Why do points made from equivalent ratios form a straight line?

PARTNER TASK

Hiking Trail, File Downloads, Street Trees. One partner writes the ordered pairs; the other plots them and checks that each point matches its table column, x first. Switch roles for each problem.

WATCH FOR

- Switching x and y, for example plotting (50, 1) for 1 hour at 50 miles per hour.
- Using uneven steps on an axis, so the points no longer line up.
- Thinking a ratio graph does not have to pass through (0, 0).

SESSION 2 · TEACHER NOTES

Interpreting Points on Ratio Graphs

TALK PROMPTS

- How does the point (1, r) show the unit rate on a ratio graph?
- How can the unit rate point help you find the pay for 11 driveways without drawing the graph?

PARTNER TASK

Water Pump, Robot Speed, Snow Shoveling. One partner reads a point aloud in words using the axis labels; the other finds the unit rate point and predicts a new point. Switch roles and check each prediction by multiplying.

WATCH FOR

- Reading a point backwards: (1, 8) as "8 items for $1."
- Dividing x by y (3 ÷ 12) instead of y by x to find the unit rate.
- Deciding a point is on the line without checking that y ÷ x matches the unit rate.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.RP.A.3a	1	1	1	1	2	2	2	3	2	1	1
S1 · Form B	6.RP.A.3a	1	1	1	1	2	2	2	3	2	1	1
S2 · Form A	6.RP.A.3a	1	1	2	2	1	2	2	3	2	1	1
S2 · Form B	6.RP.A.3a	1	1	2	2	1	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

coordinate plane: A **grid** where you plot points.

ordered pair: (x, y) names one point: **across** first, then up.

origin: The point **(0, 0)** where the axes cross.

- Each column of a ratio table becomes one **ordered pair**: x from the top row, y from the bottom row.
- Points from equivalent ratios lie on a **straight** line through the **origin**.

Example: Table: 1 tray = 8 muffins, 2 trays = **16**, 3 trays = **24**. Pairs: (1, 8), (2, 16), (3, 24). Go across x, then up y. Plot each pair, starting from (0, 0). Answer: The points lie on a **straight line** through (0, 0).

YOU TRY

1. (1, 4), (2, 8), (3, 12) on a line through (0, 0)

2. a. (2, 14) b. (5, 35)

EXIT TICKET

1. a. (2, 18) b. (4, 36)

2. Yes. Each point shows 2 for every 1, so they line up with (0, 0).

3. **B** ((0, 0))

PRACTICE FORM A

1. a. (1, 6) b. (3, 18) c. (4, 24) d. (6, 36)

2. a. 6, 9, 12 b. points (1, 3), (2, 6), (3, 9), (4, 12) on a line through (0, 0) c. (6, 18)

3. a. 15 b. 25 GB

4. a. 14, 21 b. (6, 42)

5. a. 8, 12, 20 b. y = 32 dollars c. points (1, 4), (2, 8), (3, 12), (5, 20), (8, 32) on a line through (0, 0)

6. a. y = 8x b. In 0 minutes, 0 posters are printed.

7. (3, 45), because 3 × 15 = 45 (2 × 15 = 30, 4 × 15 = 60)

8. a. A: $24, B: $30 b. Store B. Its unit rate is $2.50 per notebook, more than A's $2.00, so its cost climbs faster.

9. She switched x and y. Hours go first, so the points are (1, 50) and (2, 100).

10. **A** (x-value)

11. **B** ((3, 18))

Self-check word **ORIGIN**

PRACTICE FORM B

1. a. (1, 8) b. (3, 24) c. (5, 40) d. (7, 56)

2. a. 8, 12, 16 b. points (1, 4), (2, 8), (3, 12), (4, 16) on a line through (0, 0) c. (7, 28)

3. a. 9 b. 15 GB

4. a. 18, 27 b. (5, 45)

5. a. 10, 20, 30 b. y = 35 dollars c. points (1, 5), (2, 10), (4, 20), (6, 30), (7, 35) on a line through (0, 0)

6. a. y = 6x b. In 0 minutes, 0 posters are printed.

7. (4, 48), because 4 × 12 = 48 (3 × 12 = 36, 5 × 12 = 60)

8. a. A: $25, B: $30 b. Store B. Its unit rate is $3.00 per notebook, more than A's $2.50, so its cost climbs faster.

9. He switched x and y. Hours go first, so the points are (1, 60) and (2, 120).

10. **B** (y-value)

11. **C** ((5, 20))

Self-check word **POINTS**

GUIDED NOTES

unit rate point: The point (1, r). The r is the amount for **one**.

origin: (0, 0): zero of one quantity goes with **zero** of the other.

interpret: To say in **words** what a point means.

- A point (x, y) means x of the first quantity goes with **y** of the second. Read the axis labels.
- To find the unit rate point (1, r), divide **y** by x.

Example: Read the point: 4 pounds cost **$12**. Divide: 12 ÷ 4 = **3** dollars per pound, so the unit rate point is (1, 3). Multiply: 6 × 3 = 18. Answer: 6 pounds cost **$18**; the point is (6, 18).

YOU TRY

1. a. (1, 7) b. (1, 4.5)

2. In 3 hours, $45 is earned. 45 ÷ 3 = $15 per hour.

EXIT TICKET

1. a. (1, 7) b. (1, 1.5)

2. 2 tickets cost $10. One ticket costs 10 ÷ 2 = $5.

3. **B** (none of either quantity)

PRACTICE FORM A

1. a. (1, 5) b. (1, 9) c. (1, 2.5) d. (1, 8)

2. a. 0 bracelets need 0 charms. b. 1 bracelet needs 5 charms (the unit rate). c. 9 × 5 = 45 charms

3. a. 8 gallons per minute b. In 5 minutes, 40 gallons are pumped. c. 72 ÷ 8 = 9 minutes

4. 11 × 9 = $99

5. (8, 96): 8 × 12 = 96 meters

6. a. (1, 4) b. 10 × 4 = 40 m

7. (1, 7); yes, because 9 × 7 = 63

8. a. Rider 1: 12 km per hour; Rider 2: 10 km per hour b. Rider 1; she covers more distance each hour. c. 72 − 60 = 12 km

9. Noah read the point backwards. x is items and y is dollars, so (1, 8) means 1 item costs $8.

10. **C** (unit rate)

11. **A** (4)

Self-check word **GRAPHED**

PRACTICE FORM B

1. a. (1, 7) b. (1, 6) c. (1, 1.5) d. (1, 9)

2. a. 0 bracelets need 0 charms. b. 1 bracelet needs 4 charms (the unit rate). c. 7 × 4 = 28 charms

3. a. 6 gallons per minute b. In 5 minutes, 30 gallons are pumped. c. 48 ÷ 6 = 8 minutes

4. 7 × 12 = $84

5. (6, 66): 6 × 11 = 66 meters

6. a. (1, 5) b. 9 × 5 = 45 m

7. (1, 6); yes, because 8 × 6 = 48

8. a. Rider 1: 15 km per hour; Rider 2: 12 km per hour b. Rider 1; he covers more distance each hour. c. 75 − 60 = 15 km

9. Priya read the point backwards. x is tickets and y is dollars, so (1, 6) means 1 ticket costs $6.

10. **B** (3 boxes hold 24 pens)

11. **C** (7)

Self-check word **CHARTED**

SHEET 1 · GRAPHING RATIOS ON THE COORDINATE PLANE

1. Oranges: 3, 6, 9, 12; pairs (1, 3), (2, 6), (3, 9), (4, 12)
2. Points (1, 3), (2, 6), (3, 9), (4, 12) lie on a straight line through (0, 0). (0, 0) means 0 bags hold 0 oranges.
3. a. (7, 14) b. (9, 18)
4. a. (8, 40): 8 × 5 = 40 plants b. (0, 0): 0 rows have 0 plants
5. No. (1, 5) and (2, 10) show 5 per 1, but 3 × 5 = 15, not 14. The point (3, 14) is not on the line.
6. She switched x and y. Tickets go first: 6 tickets cost 6 × $2 = $12, so the point is (6, 12).

SHEET 2 · INTERPRETING POINTS ON RATIO GRAPHS

1. a. In 2 hours the cyclist rides 24 miles. b. 12 miles per hour c. 6 × 12 = 72 miles
2. a. 48 b. 5 c. 84 d. 3 e. 120 f. 8
3. a. 10 ÷ 4 = $2.50 per pound b. (6, 15): 15 ÷ 6 = 2.5, yes. (8, 18): 18 ÷ 8 = 2.25, no (8 lb would cost $20).
4. x is laps, so (3, 12) means 3 laps take 12 minutes (unit rate: 12 ÷ 3 = 4 minutes per lap).
5. **A** ((1, 8))
6. **A** ($9 per hour)

October 2026 conceptual additions — teacher key

Session 1 · Graphing Ratios on the Coordinate Plane

Points (0,0),(1,6),(2,12),(3,18). Horizontal: full trays; vertical: muffins. With zero trays there are zero muffins in this model.

Extension: It fits the numerical ratio but is not a permitted order of full trays. Plot discrete whole-tray possibilities for this context.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Interpreting Points on Ratio Graphs

Four pounds cost $12. (3,10) is not on this relationship because 3 pounds cost $9. Cost of 2.5 pounds is $7.50; fractional pounds are meaningful here.

Extension: Apples may be weighed in fractional pounds; a whole-tray sales rule restricts tray counts to whole numbers.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.